



**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani**  
Pilani Campus

**AUGS-AGSR Division**  
**Course Handout (Part II)**

**Date: 03/08/2026**

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

**Course Title : Supply Chain Modelling and Empirical Analysis**  
**Course Number : MF F422**  
**Instructor In-charge : SRIKANTA ROUTROY**  
**:Amit Bhomia(RS)**

**1. Scope & Objectives of the Course:**

The course provides students with the opportunity to get clarity and insight into the application of various supply chain modelling approaches and empirical analysis for evidence driven decision making in supply chain. The objectives of the course are:

- To introduce concept of decision making on the basis of multiple experts' opinion on various different supply chain problems, practices and strategies.
- To develop skill sets to analyze the qualitative data with a focus on empirical modelling for data driven decisions in supply chain.
- To develop model for life cycle analysis for measuring environmental sustainability in supply chain.
- To understand and analyse the various aspects of digital supply chain, cultural implications and their impact on supply chain performance.

**2. Text Books:**

**Text Book 1:** Hair Jr, J. F., Hult, G. T. M., Ringle, C., & Sarstedt, M. (2016), A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM). Sage publications.

**Text Book 2:** Bouchery, Yann, Charles J. Corbett, Jan C. Fransoo, and Tarkan Tan, eds. Sustainable supply chains: A research-based textbook on operations and strategy. Springer, 2016.

**3. References:**

[RB1] Hair Jr, J. F., Sarstedt, M., Ringle, C. M., & Gudergan, S. P. (2017). Advanced Issues In Partial Least Squares Structural Equation Modeling. Sage publications.

[RB2] Garson, G. D. (2016). Partial Least Squares: Regression and Structural Equation Models. Asheboro, NC: Statistical Associates Publishers.



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#### 4. Course outcomes

| No  | Course Outcomes                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| CO1 | To understand the importance of various modelling approaches and empirical analysis in supply chain.                                       |
| CO2 | To become familiar with various decision making processes considering multiple experts' opinion                                            |
| CO3 | To learn various aspects of conducting survey for empirical analysis such as questionnaire development, data collection and data cleaning. |
| CO4 | To understand and apply computer-based statistical packages for empirical analysis in supply chain.                                        |
| CO5 | To develop and analyze the supply chain model using Structural Equation Modeling approach and heterogeneous data concept.                  |
| CO6 | To understand, develop and analyze the LCA model for environmental sustainability assessment in supply chain.                              |
| CO7 | To become familiar with various aspects of digital supply chain, cultural implications and their impact on supply chain performance.       |





## 5. Course Plan

| Topic                                                     | Contents                                                                                                                                                                                                                                   | Reference                                     |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Overview of supply chain modelling and empirical analysis | Questionnaire development for experts' judgement/opinion, alternative analysis with multiple experts' opinion, cause-effect group classification, driver-dependency classification.                                                        | Research Papers and Case Studies              |
|                                                           | Contexts, issues and challenges in supply chain modelling and empirical analysis, Application in supply chain with a special focus on buyer-supplier relationship and cultural implications                                                |                                               |
|                                                           | Two case studies on applications Multi-Attribute Decision Models in supply chain.                                                                                                                                                          |                                               |
|                                                           | Learning the model development using Multi-Attribute Decision Models                                                                                                                                                                       |                                               |
| Data Collection                                           | Primary and secondary data, questionnaire design, sample size, data cleaning, Overview of Statistical Packages such as SMARTPLS3, SPSS, AMOS                                                                                               | Text Book 1, Research Papers and Case Studies |
|                                                           | Learning the different concepts of model development using Statistical Packages such as SMARTPLS3, SPSS, AMOS                                                                                                                              |                                               |
| Structural Equation Modeling                              | Measurement, Measurement Scales, Coding, Data Distributions, Structural Equation Modeling (SEM) with Partial Least Squares Path Modeling, Path Models with Latent Variables                                                                | Text Book 1, Research Papers and Case Studies |
|                                                           | Case studies and Development of Structural Equation Model                                                                                                                                                                                  |                                               |
| Modelling Heterogeneous Data                              | The Parametric Approach to PLS-Multi-Group Analysis (MGA); Measurement Invariance; Modeling Unobserved Heterogeneity; Continuous Moderator Effects.                                                                                        | Text Book 1, Research Papers and Case Studies |
|                                                           | Application of Modelling Heterogeneous Data in supply chain.                                                                                                                                                                               |                                               |
| Life Cycle Modelling                                      | Overview of Life Cycle Assessment (LCA) and its application areas, Measuring Environmental Impacts in Supply Chains, Primary and Secondary Data collection, Model Development and analysis                                                 | Text Book 2, Research Papers and Case Studies |
|                                                           | Life Cycle Assessment with model development and analysis                                                                                                                                                                                  |                                               |
| Digital supply chain and Performance modelling            | Overview of Supply chain digitalization and its impact on supply chain performance, supply chain performance measures and performance measurement system, Supply chain performance assessment, Value Stream Mapping, Cost Benefit Analysis | Research Papers and Case Studies              |
|                                                           | Recent case studies on digital supply chain, cost benefit analysis considering LCA                                                                                                                                                         |                                               |





**6. Experiential Learning Components:**

This course will feature experiential learning components in the form of projects and assignments. The students will get an opportunity to develop model for various supply chain decision making problem in various software packages such as SPSS, AMOSS, SMARTPLS3, OpenLCA and EVSM Software packages. This will also form part of the Evaluation Components for the course.

**7. Evaluation Component**

| S. No. | Evaluation Component      | Duration    | Max. Marks | Date & Time                     | Remarks              |
|--------|---------------------------|-------------|------------|---------------------------------|----------------------|
| 1.     | Mid-Semester Examination  | 90 Min.     | 50 (25 %)  | Timetable booklet               | Close Book           |
| 2.     | Class Quizzes             | TBA         | 20 (10%)   | <b>To be announced in class</b> | Close Book           |
| 3.     | Case Studies/Project      | TBA         | 60 (30%)   | <b>To be announced in class</b> |                      |
| 4.     | Comprehensive Examination | 180 minutes | 70 (35%)   | Timetable booklet               | Close plus Open Book |

**Criterion for NC:**

20% of the average of the top three performers or 30% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

**8. Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.

**Chamber Consultation:** Monday 9<sup>th</sup> Hour (4 PM to 4:50 PM]

**Notices:** All notices concerning this course will be displayed on the Mechanical Engineering Department Notice Board only.

**Make-Up Policy:** No students will be allowed to take make up without prior permission. For the case study and assignment / article/class presentation, no make ups will be granted.

**Instructor - In - charge**  
**MF F422**

